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### Method and device for delivering biological material to liquid medium

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#### Abstract

A device and method for delivering biological material and/or synthetic chemistries to a liquid medium comprising securing a capsule to an opening of a container, the capsule comprising a cap portion for securing the capsule to the container and an elongate portion comprising a cavity defined by a frangible bottom, wherein the cavity comprises live biological material therein and a plunger comprising a disrupting mechanism, suspending the elongate portion of the capsule into an interior cavity of the container, depressing the plunger, cutting or disrupting the frangible bottom of the capsule, and releasing the material into the liquid medium.

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| <b>Inventors:</b> | <b>Eviston; Mitchell (Woodbury, MN)</b>                  |
| <b>Applicant:</b> | <b>Meristem Crop Performance Group, LLC (Powell, OH)</b> |
| <b>Family ID:</b> | <b>91585070</b>                                          |
| <b>Assignee:</b>  | <b>Meristem Crop Performance Group, LLC (Powell, OH)</b> |
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*Primary Examiner:* Perry; Monica L

*Attorney, Agent or Firm:* Foley & Lardner LLP

## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This This Application is a Continuation-in-Part of U.S. patent application Ser. No. 18/503,899, filed Nov. 7, 2023, which is a Continuation of U.S. patent application Ser. No. 18/197,578, filed May 15, 2023 which is a Bypass-Continuation Application of International Application No. PCT/US2023/012628, filed Feb. 8, 2023, which claims priority to U.S. Provisional patent application Ser. No. 63/434,588, filed Dec. 22, 2022, the contents of which are hereby incorporated by reference in their entirety.

## BACKGROUND

(1) The present disclosure relates to the storing and delivery of contents to a liquid medium and more specifically to a capsule for the storage and delivery of biological material to a liquid medium for application to plants and/or soil.

(2) Farmers use soil and foliar applications for a variety of reasons including to fertilize crops, control pests and diseases, and promote crop residue breakdown. Foliar application is applying agricultural inputs (e.g., fertilizers, insecticides) directly to the leaves of a plant, not just the soil, where they can be absorbed through the leaf surface and utilized by the plant. Many applications utilize biological material (e.g., microorganisms) along with an agricultural input to promote

necessary plant processes (e.g., fertilization, pest control). However, many species of microorganisms are fragile and cannot survive long-term outside of their natural environment, thus they cannot be premixed without reducing efficacy of the agricultural input.

## SUMMARY

(3) An aspect of the present disclosure relates to a device for delivering biological material to a liquid medium. The device comprises a capsule comprising a cap portion for removably securing the capsule to a mouth of a container and an elongate portion comprising a cavity defined by a frangible bottom, wherein the cavity comprises biological material therein, and a plunger comprising a cutting or disrupting mechanism wherein the plunger is receivable within the cavity and configured to cut or disrupt the frangible bottom when activated.

(4) In a further embodiment, the device comprises a locking mechanism wherein removal of the locking mechanism exposes the plunger for activation.

(5) In one or more embodiments, the locking mechanism is a ring configured to engage a portion of the plunger opposing the cutting or disrupting mechanism.

(6) In a further embodiment, the device comprises a plunger cap to cap the portion of the plunger opposing the cutting or disrupting mechanism.

(7) In one or more embodiments, a diameter of the plunger cap is greater than a diameter of the plunger.

(8) In one or more embodiments, the portion of the plunger opposing the cutting or disrupting mechanism is configured to be engaged by a user to release the biological material stored in the cavity

(9) In one or more embodiments, the cap portion is threadably securable to the mouth of the container, thus replacing a factory cap.

(10) In one or more embodiments, the biological material comprises a bio-fungicide, bio-herbicide, bio-fertilizer, and/or bio-insecticide/pesticide.

(11) Another aspect of the present disclosure relates to a system for delivering biological material to a liquid medium. The system comprises a capsule comprising a cap portion for removably securing the capsule to a mouth of a container and an elongate portion comprising a cavity defined by a frangible bottom, wherein the cavity comprises biological material therein, a plunger comprising a cutting or disrupting mechanism wherein the plunger is receivable within the cavity and configured to cut or disrupt the frangible bottom when depressed, and the container, wherein the capsule is threadably secured to the container and wherein the elongate portion is suspended from the mouth of the container into an interior cavity of the container.

(12) In one or more embodiments, the container is a plastic F-Style container, or additional plastic liquid container for agricultural use.

(13) In one or more embodiments, a volume of the container is 2.5 gallons, 1 gallon container, pint and quart containers, drum containers, and mini-bulk tote containers.

(14) In a further embodiment, the system comprises a removable ring configured to engage a portion of the plunger opposing the cutting or disrupting mechanism and to prevent depression of the plunger.

(15) In a further embodiment, the system comprises a plunger cap configured to engage an end portion of the plunger opposing the cutting or disrupting mechanism.

(16) Another aspect of the present disclosure relates to a method for delivering biological material to a liquid medium. The method comprises securing a capsule to an opening of a container, the capsule comprising an elongate portion comprising a cavity defined by a frangible bottom, wherein the cavity comprises a plunger and biological material therein, suspending the elongate portion of the capsule into an interior cavity of the container, depressing the plunger, the plunger comprising a cutting or disrupting mechanism, cutting or disrupting the frangible bottom of the capsule, and releasing the material into the container.

(17) In a further embodiment, the method comprises removing a locking mechanism to allow

depressing the plunger.

(18) In one or more embodiments, the locking mechanism is a ring configured to engage a portion of the plunger opposing the cutting or disrupting mechanism.

(19) In a further embodiment, the method comprises engaging a plunger cap to depress the plunger, wherein the plunger cap caps an end of the plunger opposing the cutting or disrupting mechanism.

(20) In a further embodiment, the method comprises shaking the container to combine the biological material with contents provided in the container.

(21) In a further embodiment, the method comprises removing the capsule from the container.

(22) In one or more embodiments, the biological material comprises a bio-plant stimulant, bio-fungicide, bio-herbicide, bio-fertilizer, and/or bio-insecticide/pesticide.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a perspective view of a capsule for delivery of biological material according to one or more embodiments herein.

(2) FIG. 1B is a perspective view of a capsule in an activated state according to the embodiment in FIG. 1A.

(3) FIG. 2 is a cross-section view of a capsule for delivery of biological material according to one or more embodiments herein.

(4) FIG. 3A is a perspective view of a capsule installed on a container according to one or more embodiments herein.

(5) FIG. 3B is a perspective view of a capsule installed on a container according to the embodiment in FIG. 3A.

(6) FIG. 4A is a perspective view of a capsule in an activated state and installed on a container according to one or more embodiments herein.

(7) FIG. 4B is a perspective view of a capsule in an activated state and installed on a container according to the embodiment in FIG. 4A.

(8) FIG. 5A is a perspective view of a capsule body according to one or more embodiments herein.

(9) FIG. 5B is a perspective view of a capsule body according to the embodiment in FIG. 5A.

(10) FIG. 6A is a perspective view of a plunger according to one or more embodiments herein.

(11) FIG. 6B is a perspective view of a plunger according to the embodiment in FIG. 6A.

(12) FIG. 7A is a perspective view of a plunger cap according to one or more embodiments herein.

(13) FIG. 7B is a perspective view of a plunger cap according to the embodiment in FIG. 7A.

(14) FIG. 8A is a perspective view of a locking mechanism according to one or more embodiments described herein.

(15) FIG. 8B is a perspective view of a locking mechanism according to one or more embodiments described in FIG. 8A.

### DETAILED DESCRIPTION

(16) Described herein is a device for delivery of biological material to a liquid medium. The device provides a sealed, closed environment that keeps the biological material live and/or viable and disperses the biological material into a medium to be applied to plants and/or soil. The medium may be a liquid medium for spray applications such as foliar and/or soil applications.

(17) The biological material may be chemical and/or biological material. The chemical and/or biological material may be one or more chemical agents, biological agents, organisms, microorganisms, cells, viruses, proteins, compounds, molecules, etc. The chemical agents or biological agents may include fungicides, insecticides, nematicides, plant growth regulators, fertilizers, micronutrients, etc. The chemical agents are generally synthetic whereas the biological agents are generally natural (e.g., bio-fungicides, etc.). Microorganisms may include bacteria,

fungi, etc. Proteins may include enzymes, structural proteins, transport proteins, hormones, antibodies, receptor proteins, etc. The biological material includes live and/or viable biological material. For example, the live and/or viable biological material may be active, functioning, exhibiting biological processes such as growth and metabolism, capable of surviving and reproducing in its environment, etc.

(18) The liquid medium may be one or more of water, fungicides, insecticides, nematicides, plant growth regulators, fertilizers, micronutrients, compounds, molecules, etc. In one or more embodiments, the liquid medium may be a foliar and/or soil nutrient product, fungicide, insecticide, and/or herbicide. The liquid medium may be and/or comprise Excavator and/or Microbilize.

(19) Fungicides (bio-fungicides) can include: *Trichoderma* spp., *Bacillus* spp., *Pseudomonas* spp., *Streptomyces* spp., *Gliocladium* spp., *Bacillus thuringiensis* (Bt), *Coniothyrium minitans*, *Ulocladium oudemansii*, *Clonostachys rosea*, and other biological species labeled and registered as, or for, fungicidal activity.

(20) Insecticides/pesticides (bio-insecticides/pesticides) can include: *Bacillus thuringiensis* (Bt), *Beauveria bassiana*, *Metarhizium anisopliae*, *Heterorhabditis bacteriophora*, *Steinernema feltiae*, spinosad-producing bacteria, *Paecilomyces fumosoroseus*, entomopathogenic viruses, chitinolytic bacteria and fungi, *Pseudomonas fluorescens*, *Trichoderma* spp., azadirachtin-producing microbes, *Methylobacterium extorquens*, and other biological species labeled and registered as, or for, insecticidal activity. For example, *Methylobacterium extorquens* strain NLS0042.

(21) Herbicides (bio-herbicides) can include: *Phoma macrostoma*, *Colletotrichum gloeosporioides* f. sp. *aeschynomene*, *Fusarium oxysporum* f. sp. *strigae*, *Phytophthora palmivora*, *Alternaria alternata*, *Xanthomonas campestris* pv. *poae*, *Pyricularia oryzae*, *Rhopalosiphum padi* virus (RhPV), and other biological species labeled and registered as, or for, herbicidal activity.

(22) Fertilizers (bio-fertilizers) can include: *Rhizobium* spp., *Azotobacter* spp., *Azospirillum* spp., phosphate-solubilizing bacteria (PSB), mycorrhizal fungi, cyanobacteria, and *Acetobacter diazotrophicus*, *Actinomyces*, *Bacillus amyloliquefaciens*, and other biological species labeled and registered as, or for, effecting overall plant fertility activity. For example, *Bacillus amyloliquefaciens* strain ENV503.

(23) The chemical and/or biological material or the medium may include one or more of Glyphosate, 2,4-Dichlorophenoxyacetic acid (2,4-D), Dicamba, Atrazine, Paraquat, Azoxystrobin, Boscalid, Mancozeb, Myclobutanil, Thiophanate-methyl, Chlorpyrifos, Imidacloprid, Malathion, Permethrin, Deltamethrin, and other biological chemical and biological material labeled and registered as, or for, pesticidal activity.

(24) In one or more embodiments, the biological material may be from the soil, plants, or other naturally occurring biological environments. In other words, a natural environment for the biological material may be the soil, plants, and/or water. In one or more embodiments, the device is configured to mimic a natural environment of the biological material so the biological material can be stored for longer periods of time. If the biological material is added to a medium, such as a liquid medium, the biological material may only survive for a short period of time. Additionally, or alternatively, the device may keep the biological material separate from the medium. The device provides efficient and effective storage and delivery of live and/or viable biological material.

(25) Furthermore in one or more embodiments, synthetic chemistries may also be included such as conventional herbicides, fungicides, pesticide, insecticides or fertilizers.

(26) The device described herein may be a capsule that is securable to a mouth or opening of a container for dispersing biological material into the container. Live biological material includes live and/or viable biological material. The capsule may be configured to store and maintain biological material long-term and until a time for use. The capsule may contain up to about 50 mL, 125 mL, 350 mL, 500 mL, 1 L, or more of biological material therein. The capsule may be selectively deployed by a user into a container containing a medium, such as a liquid medium. The liquid

medium containing the biological material may then be applied to plants and/or the earth. The biological material may be applied to plants and/or soil for various purposes. For example, they can contribute to pest and disease control, nitrogen fixation, plant growth, weed control, bioremediation, crop residue breakdown, etc.

(27) The device comprises a capsule body having a cap portion for securing to a mouth or opening of a container and an elongate portion for storing chemical and/or biological material before deployment. In one or more embodiments, the cap portion may comprise one or more threads such that the cap portion is threadably securable to the mouth of the container. Other methods of securing the capsule body are also contemplated. The elongate portion is configured to receive a plunger therein such that the plunger may be activated by a user to disperse the chemical and/or biological material into a reservoir of the container. In one or more embodiments, the elongate portion comprises a frangible bottom such that the plunger may disrupt the frangible bottom to disperse the chemical and/or biological material into the container. The chemical and/or biological material may be stored within the elongate portion and/or the plunger.

(28) The elongate portion may be a tube or channel comprising a cavity for storage of chemical and/or biological material. The cavity may be defined by wall(s) and a frangible bottom of the elongate portion. The elongate portion may be suspended within, or above, a container when the capsule body is secured to the mouth of the container. A first end of the elongate portion located within the container may have the frangible bottom. The frangible bottom may separate contents within the capsule from contents within the container. The frangible bottom may be ruptured or broken to add the contents of the capsule to the contents of the container in the reservoir of the container. The frangible bottom may be a frangible membrane, foil, or any other material that may be disrupted by the plunger.

(29) The size and dimensions of the capsule body may vary, allowing the device to be used on any commercially available or custom designed container. The cap portion of the capsule body may be configured to screw/thread on the mouth of a container in the same manner as a factory cap of the container. For example, the cap portion may be a standard size and/or 63 mm. The elongate portion may have a diameter smaller than a diameter of the mouth of a container. A length of the elongate portion may also vary depending on the application and/or container size/volume. For example, a length of the elongate portion may be about 1-12 inches for a 1-gallon or 2.5 gallon container. A capsule volume may be proportional to a container volume. In one or more embodiments, a capsule volume is less than 25% of a container volume. For example, the capsule volume may be 10% of the container volume. In one or more embodiments, a length of the elongate portion is configured to be in contact with contents of the container such that capsule body contents are dispersed directly into the contents of the container. In an alternative embodiment, a length of the elongate portion is configured such that the capsule body is adjacent the contents of the container.

(30) The size and dimensions of a container may vary, allowing the device to be used for a variety of applications and operations. A size/volume of a container may be 2.5 gallons, 1 gallon, 1 pint, 1 quart, etc. The container may be a 125-gallon or 250-gallon tote. In one or more embodiments, a container is a jug or F-Style container. For example, the container may be a plastic 2.5-gallon F-Style container. A distinctive feature of an F-style container is its narrow mouth or opening, which is usually accompanied by a screw cap. The size and dimensions of the capsule maybe scaled up or down to accommodate larger or smaller containers. In addition, the cap portion diameter may be increased or decreased to accommodate an opening of any container.

(31) The plunger may be configured to be receivable within the elongate portion of the capsule body and may be activated to release the contents stored therein. The plunger may be one or more deployment mechanisms configured to rupture a floor or wall which acts as a dividing member separating the contents of the device from the reservoir chamber of the container. The plunger may comprise a cutting or disrupting mechanism for disrupting a frangible bottom on the capsule body allowing chemical and/or biological material to be released into the container. The disruption may



be from a blunt or a sharp element in the capsule body and/or on the plunger being displaced to displace or puncture the wall or floor of the capsule body and/or may comprise a change in the pressurization of the capsule body thus rupturing or displacing the wall(s) and/or floor of the capsule body. Other mechanisms include a plunger mechanism, puncture mechanism, slice mechanism, tearing mechanism, peeling mechanism, air pressure mechanism to effectively pop the capsule body or combinations thereof.

(32) In one or more embodiments, the device further comprises a locking mechanism to prevent premature activation of the plunger and/or deploying of chemical and/or biological material. The locking mechanism may be a ring configured to engage a portion of the plunger opposing a cutting or disrupting mechanism. Additionally, or alternatively, the locking mechanism may be located near a portion of the plunger that a user engages to activate the plunger and/or deploy capsule contents. The locking mechanism may be removed before the plunger may be activated. In one or more embodiments, removal of the locking mechanism exposes the plunger for activation by a user.

(33) In one or more embodiments, the device further comprises a plunger cap. A plunger cap may be configured for use on a portion of the plunger opposing a cutting mechanism and/or near a portion of the plunger that a user engages to activate the plunger. The plunger cap may keep a locking mechanism in place until the locking mechanism is removed by a user. Thus, the plunger cap may assist in preventing premature and/or accidental deployment of capsule contents. Additionally, or alternatively, the plunger cap may provide an easier target for a user to activate the plunger. For example, the plunger cap may have a greater diameter or area for a user to engage. The plunger cap may assist a user in selectively dispersing chemical and/or biological material into a container.

(34) Also described herein is a method of delivering chemical and/or biological material to a liquid medium. In one or more embodiments, a capsule is secured to a mouth or opening of a container. The capsule may comprise a cap portion for threadably securing to the mouth of the container and an elongate portion extending into an interior cavity of the container when the capsule is secured. The capsule may further comprise a cavity defined by the elongate portion and a frangible bottom for hosting chemical and/or biological material therein. The frangible bottom may separate the chemical and/or biological material from contents of the container.

(35) The method further comprises activating a plunger to disperse the chemical and/or biological material. In one or more embodiments, activating the plunger disrupts the frangible bottom of the capsule thereby releasing the contents therein. For example, the plunger may be activated by depressing the plunger and/or pushing the plunger toward the frangible bottom. The plunger may disrupt the frangible bottom by puncturing, popping, breaking, tearing, etc.

(36) In one or more embodiments, the method further comprises removing a locking mechanism. For example, the locking mechanism may be a ring configured to engage the plunger and prevent premature release of capsule contents.

(37) In one or more embodiments, the method further comprises engaging a plunger cap to activate the plunger. For example, the plunger cap may be engaged by depressing the plunger cap and/or pushing the plunger cap toward the plunger.

(38) In one or more embodiments, the method further comprises removing a factory cap. A container may be provided with a factory cap and the factory cap may be removed before securing the capsule to a mouth of a container.

(39) In one or more embodiments, the method further comprises shaking the container to combine the contents and/or removal of the capsule after use of the capsule. The capsule may be removed, and the factory cap may be replaced until it's time to use the mixture. At a time for use, the mixture may be applied directly to a plant and/or soil or the mixture may be combined with other products before use.

(40) A capsule **10** is illustrated in FIGS. **1-4**. The device **10** comprises a capsule body **12** having a cap portion **14** for threadably securing to a mouth or opening of a container **16** and an elongate

portion **18** for storing chemical and/or biological material **20** before deployment. The elongate portion **18** is configured to receive a plunger **22** therein such that the plunger **22** may be activated by a user to disperse the chemical and/or biological material **20** from the capsule body **12** into a reservoir of the container **16**. The elongate portion **18** and the plunger **22** collectively store chemical and/or biological material **20** therein.

(41) Referring to FIGS. **1A-1B**, the device **10** comprises a capsule body **12** having a cap portion **14** and an elongate portion **18** for storing chemical and/or biological material **20** therein (see also FIGS. **5A-5B**). FIG. **1A** illustrates a plunger **22** in a non-activated state such that the chemical and/or biological material **20** have not been released (see also FIGS. **2** and **3A-3B**). The plunger **22** is positioned within the elongate portion **18** and further comprises a plunger cap **24** and locking mechanism **26** (see also FIGS. **7A-7B** and **8A-8B**). The plunger cap **24** fits on an end of the plunger **22** opposing a cutting mechanism **28** (see also FIGS. **6A-6B**). The locking mechanism **26** is a ring configured to wrap around an end of the plunger **22** adjacent the plunger cap **24**. The locking mechanism **26** is removed before plunger **22** activation (see FIGS. **1B** and **4A-4B**). The locking mechanism **26** prevents activation of the plunger **22** before a selected time by preventing the plunger **22** and/or plunger cap **24** from being pushed and/or depressed.

(42) FIG. **1B** illustrates the plunger **22** in an activated state such that the chemical and/or biological material **20** is released into the container **16**. Referring to FIG. **1B**, the device **10** comprises a capsule body **12** having a cap portion **14** and an elongate portion **18**. The plunger **22** is in an activated and/or depressed state such that material **20** hosted within the capsule body **12** is released. The locking mechanism **26** is removed, allowing the plunger **22** to be activated. Specifically, the plunger **22** and plunger cap **24** have been pushed toward the capsule body **12** to release the chemical and/or biological material **20** into the container **16**. In the activated state, the plunger cap **24** is in contact or adjacent to the capsule body **12** and/or the elongate portion **18** of the capsule body **12**.

(43) FIG. **2** illustrates a cross-section of a capsule **10** according to one or more embodiments herein comprising chemical and/or biological material **20** therein. The device **10** comprises a capsule body **12** having a cap portion **14** and elongate portion **18**. FIG. **2** further illustrates a frangible bottom **30** on an end of the elongate portion **18** configured for suspension within a container **16** (see also FIG. **3B**). The plunger **22** is in a non-activated state such that the frangible bottom **30** is intact and the chemical and/or biological material **20** is stored within the capsule body **12** and plunger **22**. Furthermore, the device **10** comprises a plunger cap **24** and locking mechanism **26** wherein the locking mechanism **26** prevents premature activation of the plunger **22**.

(44) In one or more embodiments as illustrated in FIGS. **3-4**, a device **10** comprises a capsule body **12** having a cap portion **14** configured for threadably securing to a mouth or opening of a container **16**. In addition, the capsule body **12** comprises an elongate portion **18** for hosting a plunger **22** and chemical and/or biological material **20** therein, and a frangible bottom **30**. The frangible bottom **30** separates chemical and/or biological material **20** from contents within the container **16**. The plunger **22** comprises a cutting mechanism **28** configured for disrupting or breaking the frangible bottom **30** when the plunger **22** is activated. FIGS. **3A-3B** illustrate a device **10** in a non-activated state such that chemical and/or biological material **20** is maintained in the device **10** and the frangible bottom **30** is intact. FIGS. **4A-4B** illustrate a device **10** in an activated state such that the chemical and/or biological material **20** is released into the container **16** and the frangible bottom **30** is disrupted.

(45) Although the present disclosure has been described with reference to preferred embodiments, workers skilled in the art will recognize that changes may be made in form and detail without departing from the spirit and scope of the disclosure.

## Claims

1. A device for delivering biological material to a liquid medium comprising: a capsule comprising a cap portion for removably securing the capsule to a mouth of a container and an elongate portion comprising a cavity defined by a frangible bottom, wherein the cavity comprises biological material therein; and a plunger comprising first and second ends, wherein the first end is received within the cavity of the capsule and has a cutting or disrupting mechanism configured to cut or disrupt the frangible bottom when the plunger is depressed; wherein the second end of the plunger is exposed beyond the cavity such that the second end extends beyond the elongate portion of the capsule; and wherein the second end of the plunger is configured to interact with a locking mechanism such that removal of the locking mechanism allows the plunger to be depressed.
2. The device of claim 1 wherein the locking mechanism is a ring configured to engage a portion of the second end of the plunger.
3. The device of claim 2 further comprising a plunger cap to cap the second end of the plunger—and wherein the locking mechanism engages the second end of the plunger between the plunger cap and the elongate portion of the capsule.
4. The device of claim 3 wherein a diameter of the plunger cap is greater than a diameter of the plunger.
5. The device of claim 3 wherein the second end of the plunger is configured to be engaged by a user to release the biological material stored in the cavity.
6. The device of claim 1 wherein the cap portion is threadably securable to the mouth of the container, thus replacing a factory cap.
7. The device of claim 1 wherein the biological material comprises a bio-fungicide, bio-herbicide, bio-fertilizer, and/or bio-insecticide/pesticide.
8. The device of claim 1 wherein the container is not configured for growth and/or culture of the biological material.
9. A system for delivering biological material to a liquid medium comprising: a container comprising a mouth and an interior cavity; a capsule comprising a cap portion threadably secured to the mouth of the container and an elongate portion comprising a cavity defined by a frangible bottom, wherein the cavity comprises biological material therein, and wherein the elongate portion is suspended from the mouth of the container into the interior cavity of the container; and a plunger comprising first and second ends, wherein the first end is received within the cavity and has a cutting or disrupting mechanism configured to cut or disrupt the frangible bottom when the plunger is depressed and wherein the second end of the plunger is exposed beyond the cavity such that the second end extends beyond the elongate portion of the capsule, and wherein the second end of the plunger is configured to interact with a locking mechanism such that removal of the locking mechanism allows the plunger to be depressed.
10. The system of claim 9 wherein the container is a plastic F-Style container.
11. The system of claim 10 wherein a volume of the container is 2.5 gallons.
12. The system of claim 9 said locking mechanism comprising a removable ring configured to engage a portion of the second end of the plunger and to prevent depression of the plunger.
13. The system of claim 12 further comprising a plunger cap configured to cap the second end of the plunger and wherein the removable ring engages the second end of the plunger between the plunger cap and the elongate portion of the capsule.
14. A method for delivering biological material to a liquid medium comprising: securing a capsule to an opening of a container, the capsule comprising an elongate portion comprising a cavity defined by a frangible bottom, wherein the cavity comprises biological material therein, and a plunger having first and second ends, wherein the first end of the plunger is received within the cavity and has a cutting or disrupting mechanism and the second end of the plunger is exposed beyond the cavity such that the second end extends beyond the elongate portion of the capsule and wherein the second end of the plunger is configured to interact with a locking mechanism such that

removal of the locking mechanism allows the plunger to be depressed; suspending the elongate portion of the capsule into an interior cavity of the container; depressing the plunger, wherein depressing the plunger causes the second end of the plunger to be received within the cavity of the elongate portion; cutting or disrupting the frangible bottom of the capsule; and releasing the material into the container.

15. The method of claim 14 further comprising removing the locking mechanism to allow depressing the plunger.

16. The method of claim 15 wherein the locking mechanism is a ring configured to engage a portion of the second end of the plunger.

17. The method of claim 16 further comprising depressing a plunger cap to depress the plunger, wherein the plunger cap caps the second end of the plunger.

18. The method of claim 14 further comprising shaking the container to combine the biological material with contents provided in the container.

19. The method of claim 14 further comprising removing the capsule from the container.

20. The method of claim 14 wherein the biological material comprises a bio-fungicide, bio-herbicide, bio-fertilizer, and/or bio-insecticide/pesticide and wherein the container is not configured for growth and/or culture of the biological material.

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